

Clear Yet Insufficient: Why Users Still Return to Source Notifications After Task Extraction

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Abstract

Smartphone users often check for actionable information in notifications, yet task cues and key details can be missed during quick checks and are hard to recall after notifications are cleared. We explore whether large language models (LLMs) can extract task-related information directly from notifications and how users evaluate the extracted items in everyday use. We developed Noti, an Android notification inbox that extracts task items from users' incoming notifications. In a three-day preliminary deployment with three participants, participants rated extracted task items as correct and clear when the source notification contained a task relevant to them. However, participants still often returned to the source notification to decide next steps or verify details, especially when task context was implied in surrounding conversation or the extracted item felt irrelevant. These early findings suggest correct extraction alone may be insufficient, motivating future work on incorporating situational and user-specific context to support follow-up decisions.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**; *User studies*.

Keywords

Notification Management; Information Overload; Task Extraction

ACM Reference Format:

Uei-Dar Chen, Rui-Jie Huang, Yu-Shiun Wu, Yi-Chia Wang, and Yung-Ju Chang. 2026. Clear Yet Insufficient: Why Users Still Return to Source Notifications After Task Extraction. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, April

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ACM ISBN 979-8-4007-2281-3/26/04

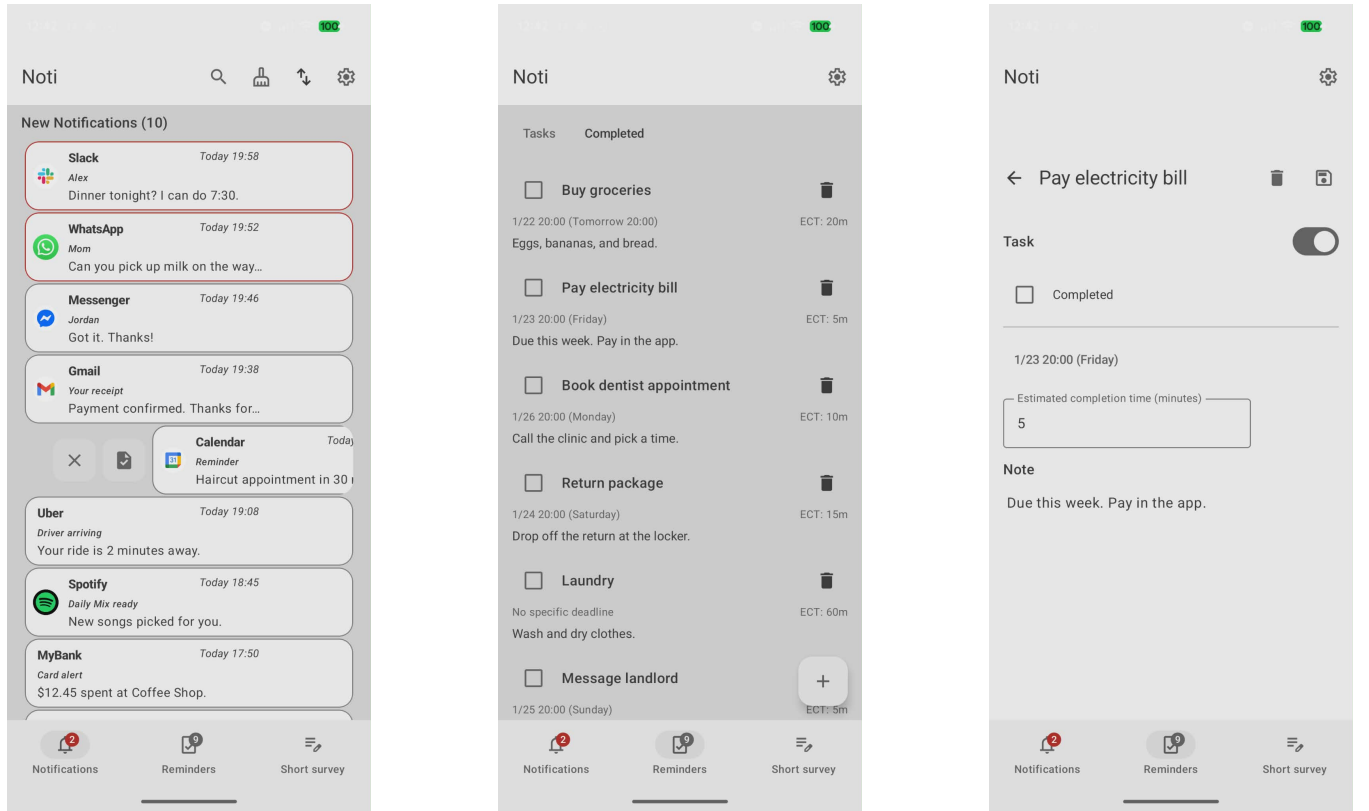
<https://doi.org/10.1145/3772363.3798322>

13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3772363.3798322>

1 Introduction

Smartphones deliver a constant stream of notifications that carry messages, updates, and alerts from many contacts and services for various daily needs [15, 17]. Research has shown that users check notifications to identify subsequent actions and determine when to act, relying on them to support task and time management [2, 3, 8, 9]. Current notification systems, however, provide limited support for extracting task-related information or preserving it in a form that supports later recall. Actionable information is often buried within long notification content, such as notifications from message threads and emails, which can be easy to miss during quick assessment [12]. Once a notification is dismissed or cleared from the notification drawer, users often need to navigate to the source app to retrieve the relevant details. This creates friction both when deciding the next step in the moment and when trying to retrieve task information later.

The ubiquitous nature of notifications in daily life and the difficulty of recovering task information motivate extracting actionable details directly from smartphone notifications. Although prior work has examined task extraction in other settings, such as deriving to-do items from emails [5, 10, 11, 19] and work communications [1, 4], notifications introduce unique challenges in practice. Notifications are often checked during short transitional breaks to support quick decision-making [2], so users have limited tolerance for false positives in extracted tasks, especially when many notifications do not lead to further actions [13, 15] or are irrelevant to the user [15, 16]. When a task does exist, details about the same obligation may be fragmented and spread across multiple notifications over time, either within an evolving thread where relevant cues are interleaved with unrelated messages, or across notifications from different sources that refer to the same obligation [3, 9]. In addition, task details can change as a thread evolves, which complicates whether an extracted task item should be created, updated, or dismissed.



(a) Notification Drawer Interface. Left-swipe reveals a task extraction button for that notification.

(b) Task List Interface

(c) Task Details Interface

Figure 1: Interface of Noti.

Recent large language models have shown strong capabilities in information extraction [6, 18], including extracting actionable information [1, 5, 7, 14], which suggests an opportunity to extract task-related information from notifications to support smartphone users’ task and time management. To explore this opportunity, we developed *Noti*, an Android application that applies large language models to users’ incoming notifications, extracts actionable information as task items, and presents them in a task list interface within the same app where users view their notifications. We evaluate Noti in a three-day preliminary deployment with three participants, combining experience sampling questionnaires and post-usage interviews to understand users’ evaluations of extracted task items and how these items fit into everyday notification interactions. Within the scope of this work, we address two preliminary research questions:

- RQ1** How accurately can LLMs extract task items from smartphone notifications?
- RQ2** How do users evaluate and interact with LLM-extracted task items from notifications in everyday use?

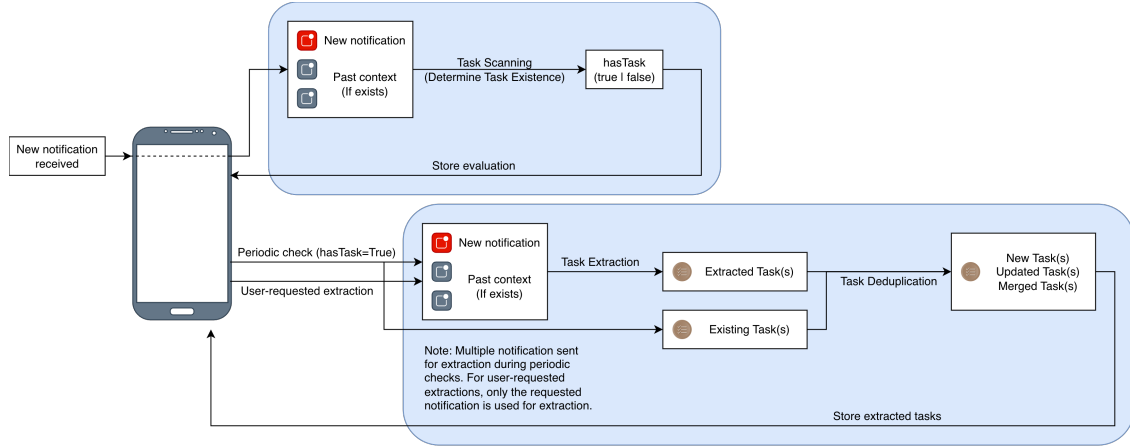
In the following sections, we describe the design of Noti and report preliminary findings from our exploratory deployment.

2 Noti: Application for Extracting Task Information From Notifications

Noti is an Android application that supports task extraction and notification management in a single interface. Noti functions as a notification inbox. It mirrors incoming notifications from the system and removes them from the native notification system, so users primarily review notifications inside Noti. Noti also offers manual features such as pinning and searching to help users manage their notification stream. Building on this unified interface, Noti uses LLMs to extract task-related actionable information from notifications and represent it as editable task items in a separate to-do list within the app. The main interfaces are shown in Figure 1, with individual screenshots explained below.

2.1 Task Extraction

2.1.1 Task Definition. We define a task to extract from notifications as *an action cue in notification content that implies the user may need to do something later*. Because notification interactions such as clicking and dismissing cannot reliably infer whether the action has been completed, a task’s pending status is treated as unknown by default and left for the user to confirm, edit, or dismiss. Based on this definition, Noti represents task-related actionable information

Figure 2: Task Extraction Workflow of Noti

as a task item. Each task item includes a title and detail text, and it may also include temporal attributes, such as an inferred deadline and a suggested completion time, when the notification content provides enough evidence to infer these values. Task items remain active until the user confirms completion or removes them.

2.1.2 Task Extraction Workflow. Noti divides task extraction into two processes, one for *scanning* whether a notification may contain a task, and another for *extracting* task details, to reduce cost and limit unnecessary task item creation. Both procedures run periodically in the background, with the detailed steps illustrated in Figure 2.

During scanning, Noti processes recent notifications and decides whether each notification may contain a task. This step reduces token cost and avoids unnecessary task generation by filtering out notifications that are unlikely to require user action. The scan uses *gemini-2.5-flash-lite* to obtain fast, low cost judgments at scale, using the notification title, body content, app name, and timestamp. When a notification belongs to a conversation or thread, such as the same message thread or email thread, Noti also includes up to five prior notifications from that thread. The scanning step outputs a binary decision that determines whether a notification should proceed to task extraction.

A notification proceeds to task extraction either when it is flagged by the scanning step or when the user requests extraction for a specific notification, as shown in Figure 1a. When a notification proceeds to task extraction, Noti uses *gemini-3-flash-preview* with a more deliberate prompting setup to generate one or more task candidates. The extraction receives the target notification and, when available, recent thread context. The extraction step produces one or more task candidates, adding temporal attributes when inferable by the notification content. Noti also preserves references to the notification keys used for extraction so users can trace each task candidate back to its source notifications.

Noti then mitigates duplicate task items and supports updates by comparing extracted candidates with the current list of active task items. This deduplication step runs in the same server call as extraction and reuses an existing task item when a candidate

overlaps in intent, rather than creating a new item. When overlap exists, Noti updates the existing task item by appending missing details and unifying notification references. When no overlap exists, Noti creates a new task item. This process also runs when the user explicitly requests extraction for a specific notification, ensuring that repeated extractions either refine an existing task item or create a new task item only when the content is independent.

2.1.3 Task Item Interface. Extracted tasks are presented in Noti’s task list view, shown in Figure 1b. Users can review extracted task items and see each item’s title, a preview of its description, and any inferred temporal attributes. Selecting a task item opens a detailed view that shows the full task content and the associated notification information, as shown in Figure 1c. Users can edit task attributes, and they can delete task items that are irrelevant or no longer needed. Users can also create task items manually for tasks that cannot be derived from notifications.

3 Evaluation Study Procedure

We invited three participants to use Noti for three days to examine how users evaluate task items extracted from notifications. These early participants were recruited via a screening form shared on social media and through word-of-mouth within the researchers’ social networks. The participants were all male, with two in their twenties and one in his fifties ($M=33.66$, $SD=14.43$). The deployment combined in-situ experience sampling questionnaires with a post-use interview to capture both immediate judgments and reflective feedback. The study protocol was approved by our institution’s IRB board, and participants received USD \$16 in compensation.

3.1 Noti Usage and ESM Assessments

We began with a 30 minute pre-study session to introduce how Noti works, explain the available features, and describe the data collected during the deployment. After consent, we helped participants install Noti and walked through the main interfaces. Participants then used Noti as their primary interface to review notifications and extracted task items throughout the three-day deployment.

We used experience sampling questionnaires to capture participants' evaluations shortly after interactions that surfaced or changed extracted task items. Each questionnaire focused on one task item and displayed both the task item and its source notification content to support recall. A questionnaire was triggered only when at least one hour had elapsed since the previous questionnaire completion and when either the participant requested task extraction for a notification or the participant viewed, edited, marked-completed, or deleted a task item. If no trigger occurred for two hours, Noti prompted a questionnaire about the most recently generated task item. To reduce disruption, participants selected a daily time window of at least twelve hours during which they were comfortable receiving questionnaires upon installing the app. Participants received at most eight questionnaires per day.

The questionnaire focused on core outcomes of task extraction from the user's perspective. It asked whether the notification contained a task the user should act upon, and whether the extracted task item helped the participant notice or confirm that task. It also asked participants to rate the correctness and clarity of the extracted task item, and whether they still needed to consult the original notification to decide next steps; when applicable, it asked why they returned to the source. When a task item included an inferred deadline or a suggested completion time, the questionnaire asked whether these values were reasonable. Unless otherwise specified, rating questions used five-point Likert scales.

During the deployment, Noti logged extracted task items together with their associated notifications, participants' interactions with task items, and experience sampling responses linked to the task items they evaluated.

3.2 Post-Usage Interviews

After the three-day deployment, we conducted a semi-structured interview of 30 to 60 minutes with each participant. We used their experience sampling responses as probes and asked how task extraction and task items fit into their notification routines, what benefits they experienced in retrieving task-related information, and what limitations they encountered. We also asked when participants chose to edit or delete task items, when they preferred to return to the original notification, and what changes would make extracted task items more useful and reliable.

4 Preliminary Results

4.1 Quantitative Results

Across the three-day deployment, participants generated an average of 19.67 extracted task items per day ($SD=7.32$). This count includes both newly created task items and updates to existing task items triggered by later notifications within the same task context. Among these extractions, 95.73% were initiated automatically after the scanning step flagged a notification as likely containing a task, while 4.27% were initiated by user requests.

Participants collectively completed 34 experience sampling questionnaires. In these ESM-evaluated cases, participants reported that the source notification mentioned a task they should act upon in two thirds of questionnaires (64.71%). When a user-relevant task was present, participants reported that the extracted task item helped them notice or confirm a pending item ($M=4.33$, $SD=0.61$) and

rated the extracted description positively in correctness ($M=4.42$, $SD=0.52$) and clarity ($M=4.00$, $SD=0.36$). Conversely, in 35.29% of ESM-evaluated cases, participants reported that the source notification did not contain a task they should act upon, indicating false positives from the user's perspective.

Despite positive ratings when tasks were present, participants still reported needing or being unsure whether they needed to return to the source notification for 42.86% of extracted items they judged to be tasks. Among cases where they returned or were unsure, the most frequently selected reasons were that the task item did not provide enough information to decide the next step (55.56%) and that participants wanted to verify the original wording or details to avoid misunderstanding (55.56%).

4.2 Qualitative Findings

Participants generally described extracted task items as helpful for reducing effort in locating actionable information within a high volume of notifications. They noted that task-related notifications are often interleaved with non-action content in the notification drawer, and that viewing extracted task items in a separate list helped them identify potential tasks without reading through unrelated content, allowing them to “*read through less junk*” (P1). Participants also appreciated cases where Noti consolidated updates from multiple related notifications into a single task item as details accumulated over time. P1 explained, “*When something's broken and a report on Slack hasn't been addressed, messages could start flooding in from Messenger, Telegram, [...], and in that case, I just want one task telling me 'it's broken, please fix it.'*”

At the same time, even when extracted items were perceived as helpful for surfacing actions, participants often returned to source notifications to reduce the risk of misunderstanding, aligning with questionnaire responses. They described returning to recover context that was not captured in the extracted task item, such as urgency and rationale, particularly in informal conversations. They noted that although not every task includes an explicit deadline, surrounding conversation can provide cues such as “*Do this early*” or “*It's urgent*” that help them assess priority. Participants also emphasized that understanding *why* a step is required can signal additional follow-up work beyond the explicitly assigned action. P2 explained, “*During work we will mention a reason for doing parts of the procedure, and I'll know the following steps that also has to be done. Now the LLM just tells me the steps that others assigned me to do, but if I don't know why, I might miss subsequent steps I should also work on as well.*”

Finally, participants noted that extracting an action without sufficient situational context can surface items they do not consider to be *their* tasks, especially in group conversations where responsibility is implied rather than explicitly stated. They cautioned that such extractions could add burden and undermine the goal of making relevant task information easier to access. This concern is consistent with the experience sampling results, where participants reported that the evaluated notification did not contain a task they should act upon in about one-third of questionnaires. P2 elaborated, “*Someone in the chat tagged another person [in the group] and told him to do something, but the LLM simply extracted the task and treated it as something that I should be doing, but I don't need to know about*

that.” Participants also suggested that task relevance depends on user-specific preferences and context. P1 explained, “I should be able to say, ‘This [task item] is from an ad,’ and I don’t want it.”

Overall, this preliminary deployment suggests that LLM-extracted task items may reduce effort in spotting potential actions, especially when actionable details are scattered across multiple updates. However, participants in this pilot still relied on source notifications to recover missing context and to judge task relevance, highlighting the potential importance of situational and user-specific cues beyond the extracted action itself.

5 Conclusion and Future Work

This work explores task extraction directly from smartphone notifications and offers preliminary evidence about its potential benefits and limits in everyday use. We developed Noti, an Android application that uses large language models to extract task-related actionable information from incoming notifications and represent it as task items alongside an in-app notification drawer. In a three-day preliminary deployment with three participants, extracted task items helped participants locate potential actions without repeatedly scanning the full stack of notifications, particularly when task cues were scattered across multiple updates.

At the same time, our findings from this preliminary deployment suggest that extracting an action alone may be insufficient for supporting follow-up decisions. Participants often returned to source notifications to recover context not captured in task items and to assess whether an extracted action was actually relevant to them. In future work, we will examine what contextual cues users need alongside extracted actions to support prioritization and next-step decisions. An open question is what kinds of context should be surfaced, and in which situations, so that task items can better support follow-up without requiring users to return to the source notification. We will also explore how incorporating more context from notification threads and users’ personal context could reduce irrelevant extractions. Together, these directions aim to move from correct extraction to task items that users can trust and act on in everyday life.

Acknowledgments

We sincerely thank all study participants and appreciate the support of the National Science Technology Council, Taiwan (113-2222-E-A49-009-MY3).

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